

Forecasting European Housing Price Growth

Using Machine Learning and Housing Market Regime Clustering

Authors

Carla de Erausquin (185899), Christoph Schilling (185939), Giulia Ferrari (185877),

Copenhagen Business School - MSc Business Administration and Data Science
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Abstract

This paper forecasts one-quarter-ahead residential house price growth across 26 EU member states using a quarterly panel of 1,704 observations (2005–2025). We compare benchmark models, pooled ML models, and two cluster-stratified variants based on data-driven K-Means and theory-grounded West/North/South/East housing market regimes, evaluated on a held-out 2022–2025 test set. Pooled ML models achieve modest improvements over naïve persistence (Ridge: -12.1%), with regularised linear models outperforming tree-based methods. Literature-based stratification substantially outperforms the pooled baseline for North ($+29.5\%$), West ($+14.3\%$), and East ($+11.9\%$) clusters; K-Means yields mixed results. Feature importance confirms long-term interest rates and autoregressive HPI lags as dominant predictors, with meaningful variation across regimes. These results suggest that model design (particularly stratification scheme) matters more than complexity, and that institutional knowledge should guide cluster design in multi-country forecasting.

Keywords: *House Price Forecasting, EU Housing Market, Machine Learning, Panel Data, Unsupervised Learning, Housing Market Regimes, Interest Rates, Gradient Boosting, K-Means.*

1 Introduction

Residential housing markets are central to household wealth, financial stability, and macroeconomic policy. Despite their importance, short-horizon forecasting of house price growth remains challenging, particularly across heterogeneous multi-country panels where structural differences between markets may cause the same macroeconomic inputs to produce different outcomes. This paper asks whether combining machine learning forecasting with explicit housing market regime clustering can improve

one-quarter-ahead predictions of house price growth across 26 EU member states (EU-26, i.e., EU-27 excluding Greece due to data unavailability). We address three research questions: **(RQ1)** Can ML models outperform naïve and linear benchmarks? **(RQ2)** Does stratifying by housing market regime improve accuracy over a pooled model? **(RQ3)** Which macroeconomic and institutional factors most strongly drive short-term housing price dynamics, and do these vary across regimes?

We contribute a quarterly EU-26 panel framework that directly compares data-driven clustering (PCA + K-Means + DBSCAN) against a theory-grounded grouping (West/North/South/East), and evaluates both against a pooled baseline on a held-out 2022–2025 test period. Sections 2–8 cover related work, the conceptual framework, methodology, results, ethical considerations, and conclusions.

2 Related Work

2.1 House Price Determinants and Forecasting

Egert & Mihaljek (2007) analyze the determinants of house price dynamics in Central and Eastern European countries (CEECs), highlighting that housing markets in transition economies differ significantly from those in more mature OECD economies. They show that macroeconomic and demographic factors affect house prices differently across countries, motivating their distinction between CEECs and more advanced economies.

Consistent with this, Andrews et al. (2011) demonstrate that housing markets across OECD countries vary systematically along structural dimensions (e.g., supply responsiveness, tenure structure, credit access, and regulatory frameworks) influencing house price levels, volatility, and broader outcomes such as residential mobility.

Building on this heterogeneity, Calza, et al. (2013) show that differences in mortgage market structures across European countries such as loan-to-value ratios, availability of equity withdrawal, and especially the prevalence of fixed versus variable interest rate contracts lead to significant variation in how monetary policy affects house prices, residential investment, and consumption. In particular, countries

with more flexible mortgage markets and variable-rate structures exhibit stronger sensitivity to interest rate changes.

2.2 Machine Learning in Macroeconomic Forecasting

Machine learning methods have shown promise in macroeconomic forecasting settings where relationships between variables are non-linear or unstable over time. Medeiros et al. (2021) demonstrate that gradient boosting and random forest models outperform linear benchmarks for inflation forecasting in a high-dimensional data environment, particularly during periods of structural change.

Similarly, Goulet Coulombe et al. (2022) find that ML models provide the largest forecasting gains over linear models precisely when the economy is in a distinct regime, a finding that directly motivates our cluster-stratified modelling approach.

However, most existing applications focus on single-country or aggregate-level data. This paper contributes a quarterly EU-26 panel framework that combines supervised ML forecasting with an explicit comparison of data-driven and theory-based regime clustering, an approach that has received limited attention in the literature to European housing price dynamics.

3 Conceptual Framework

3.1 Data Preparation

We combine multi-source panel data from 2005–2025 across EU-26 countries, process it into a quarterly panel, and feed it through a four-phase modelling pipeline.

3.2 Analytical Pipeline

This study employs a panel time series dataset of 1,704 quarterly observations across 26 EU countries spanning 2005 to 2025. The dataset is split temporally: all observations from before 2022 form the training set ($n=1,340$), and all observations from 2022 onwards form the held-out test set ($n=364$), as seen on *Figure 1*. The test set is not used at any stage of model training or hyperparameter selection; it is reserved exclusively for final evaluation, ensuring that reported performance reflects genuine out-of-sample forecasting ability.

Because the data has a panel time series structure, standard random k-fold cross-validation would constitute data leakage: a model trained partly on 2020 data to predict 2018 prices would be learning from the future. Instead, we apply walk-forward expanding-window cross-validation within the training set, implemented via `TimeSeriesSplit` with five folds. In each fold, the training window expands by one step and the model is validated on the immediately following period, always training on the past and validating on the future, mirroring real-world forecasting conditions (Géron, 2022, Ch. 2). Cross-validation is used solely for hyperparameter selection; final model performance is measured only on the test set.

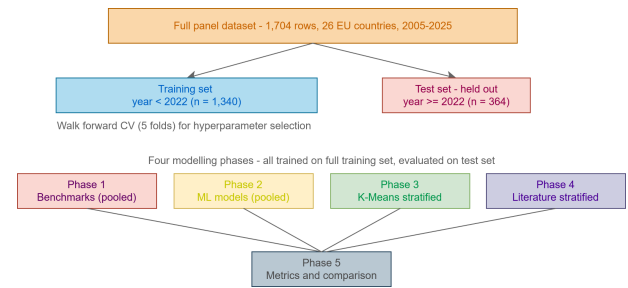


Figure 1: Overview of the modeling pipeline, with the already processed dataset.

The analysis proceeds through four phases, all of which use the same training and test split. *Phase 1* establishes performance benchmarks using naive persistence, rolling mean, and OLS regression. *Phase 2* trains regularised linear models (Ridge, Lasso) and tree-based ML models (Random Forest, Gradient Boosting) on the full pooled dataset. *Phase 3* and *Phase 4* repeat the *Phase 2* models but trained separately within each housing market cluster (*Phase 3* using data-driven K-Means clusters and *Phase 4* using the theory-based literature clusters). All predictions are collected and compared in *Phase 5* using three metrics: RMSE, MAE, and R^2 as seen on *Figure 1*.

3.3 Clustering as a Modelling Strategy

The central motivation for clustering is that housing markets in different structural regimes respond differently to the same macroeconomic inputs. If rising interest rates suppress demand more strongly in variable-rate mortgage markets (Calza et al., 2013; Egert & Mihaljek, 2007), a pooled model trained across mixed institutional contexts will fail to capture either relationship precisely. Regime-specific models, trained only on structurally similar countries, can learn more precise relationships and produce better forecasts.

We compare two clustering schemes. The data-driven scheme (PCA + K-Means, $K=4$) identifies groupings that emerge naturally from country structural

profiles, without imposing prior theory. The literature-based scheme (West, North, South, East; Andrews et al., 2011; Egert & Mihaljek, 2007) tests whether theoretically motivated groupings produce better forecasting stratification. Ireland is assigned to the East cluster on the basis of its documented institutional transition from a state-managed to a marketised housing regime, coinciding with the post-communist transitions of Eastern Europe, and its historically high price volatility (Norris, 2016; Stephens et al., 2015), rather than geographic location.

4 Methodology

4.1 Dataset Description

The dataset combines macroeconomic, financial, demographic, and political variables to capture the main theoretical drivers of housing price dynamics (Table 1). The House Price Index (HPI) serves as the dependent variable, reflecting residential price movements across countries. On the demand side, HICP inflation captures cost pressures and purchasing power effects, while GDP and unemployment rates proxy for economic activity and labour market conditions influencing housing demand. On the supply side, building permits are included as a forward-looking indicator of housing construction.

Financial conditions are captured through mortgage interest rates (MIR) and long-term sovereign yields. Demographic variables (old-age dependency ratio and population density) control for structural cross-country differences, while political ideology (RILE) accounts for institutional and policy variation across markets.

Dataset	Source	Frequency	Role
House Price Index (HPI)	Eurostat <i>prc_hpi_q</i>	Quarterly	Target variable
HICP Inflation	Eurostat <i>prc_hicp_midx</i>	Monthly → Quarterly	Demand-side cost pressure
GDP Index	Eurostat <i>namq_10_gdp</i>	Quarterly	Economic activity
Unemployment Rate	Eurostat <i>une_rt_q</i>	Quarterly	Labour market conditions
Building Permits	Eurostat <i>sts_cobp_q</i>	Quarterly	Housing supply proxy
Old-age Dependency Ratio	Eurostat <i>demo_pjanind</i>	Annual → Quarterly	Structural demographics
Population Density	Eurostat <i>demo_r_d3dens</i>	Annual → Quarterly	Structural demographics
MIR Mortgage Rates	ECB API (MIR)	Monthly → Quarterly	Cost of borrowing (eurozone)
Long-term Interest Rates	Eurostat <i>irt_lt_mcby_q</i>	Quarterly	Sovereign yields (all EU-27)
Parliament Ideology (rile)	MARPOR 2025a	Per election → Quarterly	Institutional/political orientation

Table 1: Sub-datasets extracted for the final training dataset creation.

The final dataset is an unbalanced quarterly panel comprising 1,704 observations across 26 EU countries (EU-27 excluding Greece due to data unavailability) over the period 2005Q3–2025Q2, with a total of 39 variables after preprocessing and feature construction, as we can see on Figure 2. The panel then combines macroeconomic, financial, demographic, and political indicators from multiple sources, harmonized at the country–quarter level.

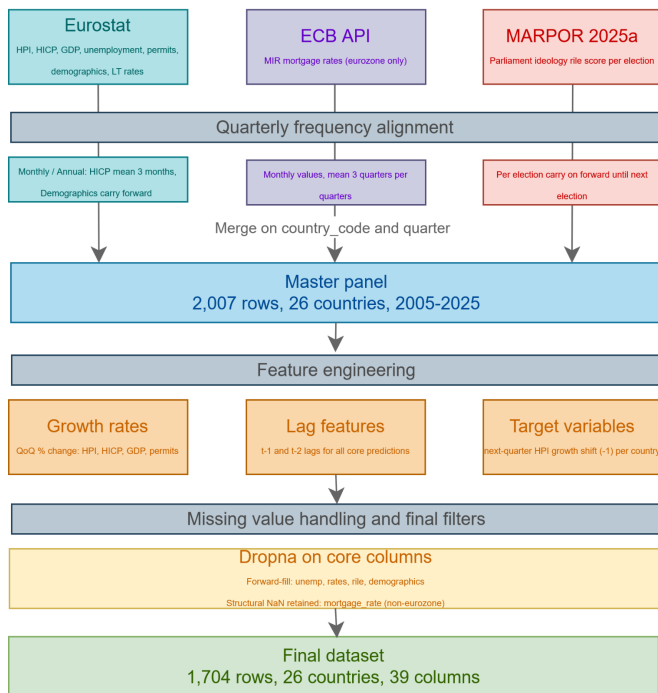


Figure 2: Data preparation process.

Regarding missing data, different strategies are applied depending on the nature of the variable. Long-term interest rates are used as a universal proxy for borrowing costs across all countries. In contrast, mortgage rates (MIR) are only available for eurozone countries; missing values for non-eurozone countries are therefore treated as structural missingness rather than imputed. Political ideology scores (RILE) from MARPOR are forward-filled between elections, reflecting the persistence of parliamentary composition over time. Standard forward-filling is also applied to low-frequency structural variables such as demographics.

4.2 Data Pre-processing

4.2.1 Frequency Alignment

All variables are converted to a quarterly frequency to ensure temporal consistency. Monthly variables, such as HICP inflation and MIR mortgage rates, are aggregated using simple quarterly averages.

Annual variables, including demographic indicators, are carried forward within the year. Political variables from MARPOR, observed at election dates, are carried forward until the next election, capturing institutional persistence.

4.2.2 Feature Engineering

To ensure stationarity and comparability across countries, quarter-on-quarter growth rates are computed for key macroeconomic variables, including HPI, HICP, GDP, and building permits. The target variable is defined as next-quarter HPI growth, obtained by shifting the dependent variable one period ahead.

To capture short-term dynamics, two lags ($t-1$ and $t-2$) are included for all core predictors. Additionally, a time-varying eurozone membership indicator is incorporated to account for differences in monetary regime and data availability, particularly for mortgage rates.

4.3 Exploratory Data Analysis

The first analyzed variable of the dataset was the target variable, HPI next quarter values (`target_hpi_next_q`). Understanding its distribution, temporal behaviour and cross-country variation was the priority of the EDA. As seen on Figure 3, the HPI next quarter growth exhibits a near normal distribution, symmetric and with moderate tails.

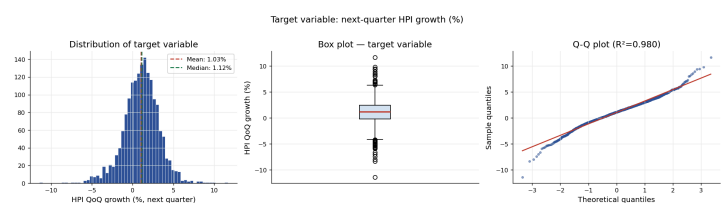


Figure 3 (a,b,c): Target distribution, near-normal (skewness -0.24, kurtosis 1.94), 70.7% positive quarters).

Regarding temporal dependence, *Figure 4* suggests that HPI growth exhibits moderate short-term persistence, with significant positive autocorrelation up to *lag 4*. The gradually declining ACF and the strong first-lag PACF indicate that recent past values contain relevant predictive information, supporting the inclusion of lagged features ($t-1$ and $t-2$) in the model.

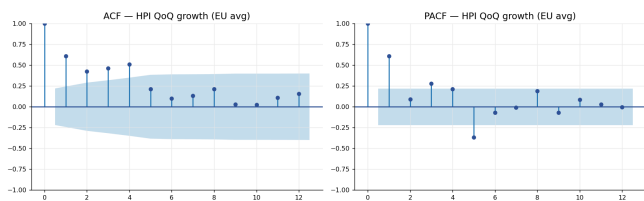


Figure 4: ACF and PACF of target variable (lag-1: 0.632, lag-2: 0.445, lag-3: 0.490, lag-4: 0.548)

The correlation matrix in *Figure 5* reveals one notable interpredictor relationship: *lt_interest_rate* and *mortgage_rate_pct* are correlated at $r=0.67$, reflecting the tendency of retail mortgage pricing to track sovereign bond yields. This collinearity is addressed in the feature selection stage by retaining only *lt_interest_rate* as the universal interest rate proxy across all 26 countries, given that *mortgage_rate_pct* is structurally missing for non-eurozone members. All remaining predictor pairs show correlations below 0.40, indicating that multicollinearity is not a concern for the linear models in this study.

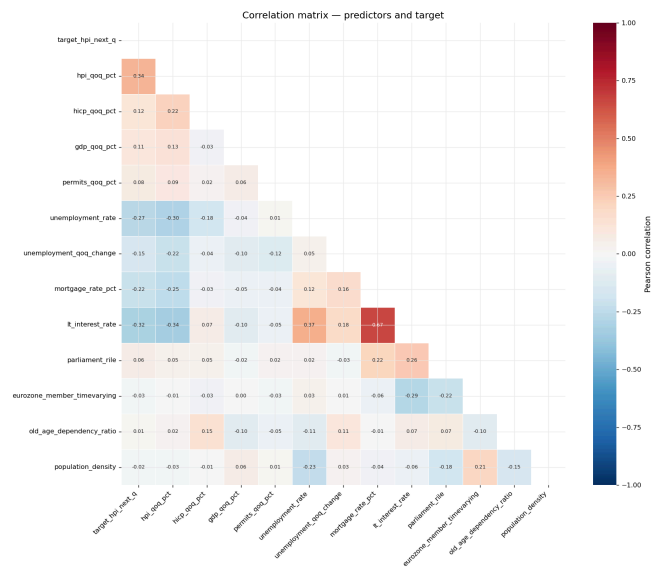


Figure 5: Correlation matrix between predictors and target.

As *Figure 6* shows, price growth volatility is highly heterogeneous across the EU-26, with Eastern and marginal markets (e.g., Cyprus, Latvia, and Hungary) far exceeding the EU mean, while Western European markets like Belgium and France cluster well below.



Figure 6: Price growth volatility by country. Eastern markets (HU, EE, LT) average 2x growth of Southern Markets (IT, FI).

The panel is unbalanced in its early years, with France and Sweden providing observations from 2005 while most countries enter the sample from 2008 onward, reflecting differences in Eurostat data availability. As a result, the pre-2008 quarters are represented by a limited subset of predominantly Western European markets, which should be kept in mind when interpreting cross-country comparisons for that period. Furthermore, the 2008-2012 period

captures the Global Financial Crisis and European sovereign debt crisis which are extreme but economically real events that provide important signals for recognising downside regimes.

Figure 7 presents a heatmap of mean quarterly HPI growth by country and year. The 2008–2012 crisis appears as a band of contractions across most markets, sharpest in Ireland, Spain, and the Baltics. The post-2015 recovery is broadly shared, though Eastern markets show stronger growth. The 2021–2022 boom and 2023 correction represent the structural break that makes the test period challenging.

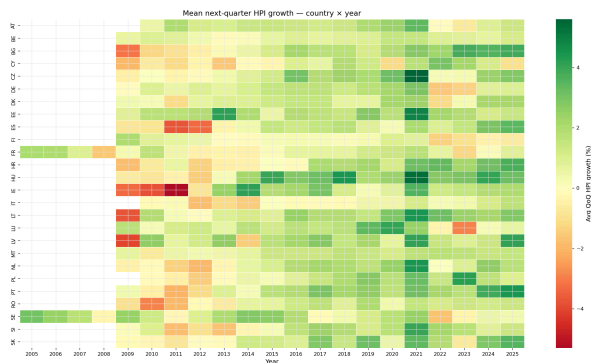


Figure 7: Mean quarterly HPI growth by country and year.

In order to choose *when* to choose the train-test split, the observations per year were assessed, as seen on the Appendix (Figure A2). Based on this analysis, the data was split into two, as seen on Figure 1: training data being the data prior to the year 2022 and test data, being 2022 forward, being the latter 21% of the whole dataset.

The distribution of the target variable is broadly symmetric within most countries, with median quarterly growth positive across 24 of 26 markets. However, substantial cross-country heterogeneity is evident in both the central tendency and spread of growth rates. Estonia, Hungary, and Lithuania show the highest median growth, while Italy, Finland, and

France cluster near to zero. Cyprus is a notable outlier with an exceptionally wide interquartile range reflecting its boom-bust cycle, as seen on the Appendix (Figure A1). This within-country symmetry is consistent with the near-normal pooled distribution reported in Figure 3, though the variation in spread across countries motivates the cluster-stratified modelling approach.

4.4 Clustering Analysis

4.4.1 Literature-Based Clustering

Literature-based clustering was performed, as seen on Table 2. Simple assessments were made on these clusters, including ANOVA, with $F = 7.56$ and $p < 0.001$. Therefore, there is strong evidence to claim that the means of these clusters differ.

Group	Countries	Key Literature Rationale
West	AT, BE, DE, FR, LU, NL	Coordinated Market Economies (Hall & Soskice, 2001); low volatility confirmed in EDA (BE std=1.07, FR std=1.44)
North	DK, FI, SE	Nordic welfare states with variable-rate mortgage traditions (Andrews et al., 2011); among most stable markets in EDA
South	CY, ES, IT, MT, PT	Mediterranean; high homeownership and boom-bust dynamics 2008-2014 (Andrews et al., 2011); internal heterogeneity confirmed in EDA
East	BG, CZ, EE, HR, HU, IE, LT, LV, PL, RO, SI, SK	Post-communist convergence and institutional transition (Stephens 2015, Egert & Mihaljek 2007, Ireland specifically: Norris 2016); highest avg growth and volatility in EDA

Table 2: Literature based clustering.

4.4.2 Data-Driven Clustering (PCA + K-Means + DBSCAN)

Two other different clustering approaches were conducted. Both K-Means and DBSCAN on the dimensionality reduced dataset, performed with PCA. The reason for two additional clusterings is straightforward: K-Means assumes clusters are spherical and similar in size. DBSCAN makes no such

assumptions. More importantly, DBSCAN explicitly identifies outlier countries that do not fit any cluster.

Country structural profiles for K-Means clustering were computed exclusively from the training window (2005-Q3 to 2021-Q4), ensuring that no test-period information entered the cluster assignments.

For choosing the amount of components before both clusterings, as seen on *Figure 8*, the variance retained was analyzed. Keeping four components explained 82.0% of the variance, therefore used on the following steps.

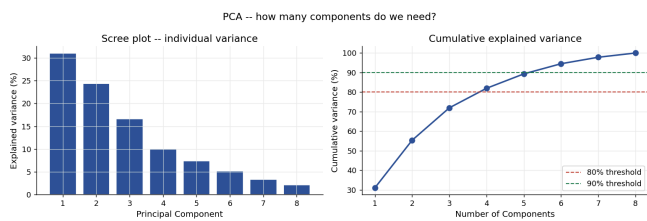


Figure 8: Variance retained by each PCA component.

The PCA biplot in *Figure 9* reveals two dominant structural axes. PC1 (31.0% variance) separates high-growth, volatile Eastern markets (HU, EE, LV) from stagnant Western and Southern markets (FI, IT, FR) along a dynamism dimension driven by HPI volatility and target growth. PC2 (24.3% variance) reflects a political orientation and density axis, with parliament ideology and long-term interest rates pointing toward Eastern markets and population density toward the structural outliers Malta and Luxembourg.

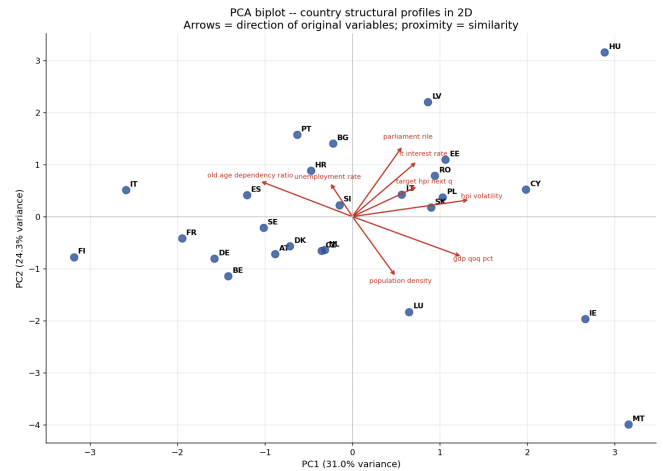


Figure 9: PCA biplot.

To determine the optimal number of clusters, two complementary methods were applied: the elbow method and the silhouette score, as shown in *Figure 10*. The elbow method plots within-cluster inertia against K ; the rate of improvement decelerates noticeably after $K=4$, suggesting that additional clusters beyond this point offer diminishing returns in compactness.

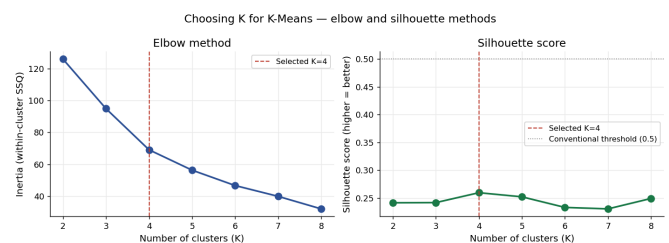


Figure 10: Elbow method and Silhouette scores for $K \in [2,8]$.

Given the elbow at $K=4$ and the modest peak at $K = 4$, this was the number of clusters selected, as seen on *Figure 11*. *Table 3* and *Figure A4* better display the different components that played a more active role in the clustering.

Group	Countries	Empirical Profile (see Figure A4)
0 Stable Growth	AT, BE, CZ, DE, DK, EE, LU, NL, SE, SI	Above-average HPI growth (+1.07 SD), lowest interest rates (−1.25 SD), low unemployment. Institutionally stable, Low-rate markets with moderate-to-strong price appreciation.
1 Stagnation	BG, ES, FI, FR, HR, IT, PT	Lowest HPI growth (−1.23 SD), highest unemployment (+1.30 SD), high old-age dependency (+1.24 SD), lowest volatility (−1.02 SD). Structurally ageing, high-unemployment markets with stagnant price dynamics.
2 Structural Outliers	IE, MT	Highest population density (+1.49 SD) and GDP growth (+1.44 SD), highest volatility (+1.23 SD), most right-leaning parliament (−1.06 SD on RILE scale). Structural outliers — Ireland and Malta share extreme values on different dimensions rather than genuine regime similarity.
3 Conver- gence	CY, HU, LT, LV, PL, RO, SK	Highest interest rates (+1.18 SD) and above-average HPI growth (+0.48 SD), right-leaning political orientation (+1.35 SD). High-rate, post-transition convergence economies with above-average growth momentum.

Table 3: K-Means based clustering.

The silhouette scores, while below conventional thresholds designed for well-separated datasets, do not invalidate this choice. Rather, they confirm that EU housing markets form a structural continuum rather than discrete regimes, a finding subsequently corroborated by the DBSCAN analysis.

DBSCAN was applied to the PCA-reduced country profiles across three epsilon values (1.5, 1.75, 2.0). In all cases the algorithm identified a single dense cluster containing the majority of the analyzed countries, with a small set of noise points (ES, HU, IE, IT, MT), confirming their status as structural outliers that do not fit any natural density-based grouping, as seen on *Figure A3*. This result indicates that European housing markets do not form tight, well-separated clusters in feature space. Rather, most countries lie on a continuum with a handful of extreme cases at the periphery.

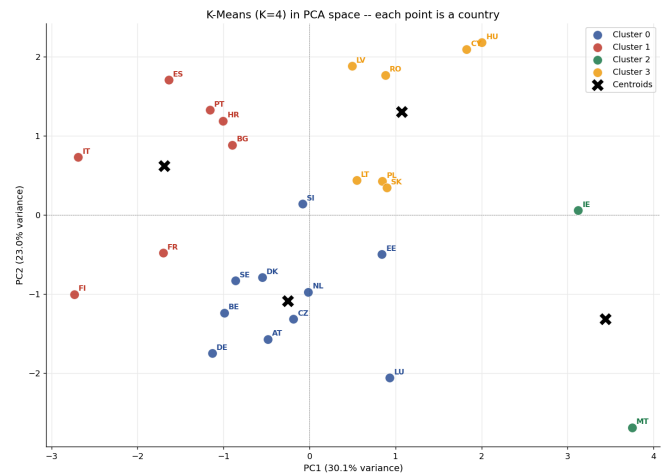


Figure 11: K-Means clustering results, on PCA space.

As seen on *Figure 11*, Finland sits close to Italy rather than its Nordic peers, supporting its K-Means classification into the Southern stagnation cluster; Portugal is displaced from the Mediterranean group toward the Eastern convergence markets; and Ireland and Malta are isolated at the periphery of the space, consistent with their persistent classification as noise points in the DBSCAN analysis.

Consequently, the K-Means and literature-based clusterings are retained as the primary stratification schemes for the modelling phase.

4.5 Modelling Framework

4.5.1 Benchmark Models

Three baseline models are used to establish reference performance levels.

First, a naïve persistence model predicts next-quarter growth (y_{t+1}) as the current value (y_t):

$$y_{t+1} = y_t$$

providing a minimum benchmark that any predictive model should outperform.

Second, a rolling mean model predicts the average of the previous four quarters. This

specification is motivated by the autocorrelation structure identified in the exploratory analysis, particularly the significant lag-4 autocorrelation, suggesting short-term cyclical dynamics:

$$y_{t+1} = \frac{1}{4} \sum_{i=0}^3 y_{t-i}$$

Finally, an ordinary least squares (OLS) regression is implemented as the simplest parametric model:

$$y_{t+1} = \beta_0 + \sum_{k=1}^K \beta_k x_{k,t} + \varepsilon_t$$

β_0 being the intercept, β_k the coefficients associated with each predictor $x_{k,t}$, and ε_t the error term capturing the part of housing price growth not explained by the model. Intuitively, the coefficients β_k measure the marginal effect of each variable on next-quarter house price growth, holding other variables constant.

This benchmark evaluates the extent to which linear relationships alone can explain housing price dynamics before introducing regularisation or non-linear methods.

4.5.2 Regularised Linear Models

To address potential multicollinearity and overfitting in the high-dimensional feature space, regularised linear models are employed.

A Ridge regression (L2 regularisation) penalises large coefficients by shrinking them toward zero without eliminating variables. The regularisation parameter is selected via cross-validation using *RidgeCV*.

$$\min_{\beta} \left[\sum_{i=1}^n (y_i - \hat{y}_i)^2 + \lambda \sum_{k=1}^K \beta_k^2 \right]$$

Unlike OLS, regularised models introduce a penalty term controlled by λ which constrains coefficient magnitudes and reduces overfitting. It controls how much the model penalises large coefficients.

A Lasso regression (L1 regularisation) introduces sparsity by shrinking some coefficients exactly to zero, effectively performing feature selection. The optimal penalty is determined via *LassoCV*:

$$\min_{\beta} \left[\sum_{i=1}^n (y_i - \hat{y}_i)^2 + \lambda \sum_{k=1}^K |\beta_k| \right]$$

In the final specification, several lagged variables are eliminated, including unemployment and permit building lags and long-term interest rate lags, indicating limited marginal predictive contribution.

4.5.3 Tree-Based Models

To capture non-linear relationships and interactions between variables, two ensemble tree-based methods are implemented.

A Random Forest constructs an ensemble of decision trees on bootstrapped samples, reducing variance through averaging; the best-performing configuration uses 300 trees, unrestricted depth, and a minimum of five observations per leaf.

Gradient Boosting builds trees sequentially, each correcting the residual errors of the previous; the optimal configuration includes 200 estimators, a learning rate of 0.05, maximum depth of 3, and subsampling of 0.8.

4.5.4 Walk-Forward Cross-Validation

Given the temporal structure of the data, standard random cross-validation would lead to information leakage. Instead, a walk-forward

(expanding window) cross-validation strategy is employed, as seen on *Figure 12*.

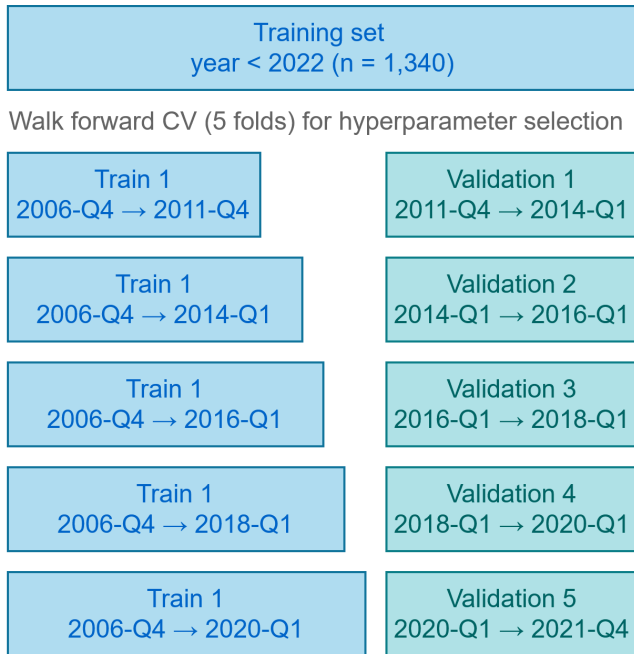


Figure 12: Walk-forward Cross-Validation schema for hyperparameter selection of training data.

The training window expands sequentially across five folds, each predicting one quarter ahead, ensuring all validation sets strictly follow the training period and preserving realistic out-of-sample evaluation. Implementation is carried out using *TimeSeriesSplit* from *scikit-learn*.

4.6 Evaluation Metrics

Model performance is evaluated using three complementary metrics.

The primary metric is Root Mean Squared Error (RMSE), which penalises large prediction errors more heavily. This is particularly relevant in the context of housing markets, where extreme movements (e.g., during crises or booms) carry significant economic implications.

$$RMSE = \sqrt{\frac{\sum_{i=1}^n (\hat{y}_i - y_i)^2}{n}}$$

y_i = value of observation i
 $\hat{y}$ = predicted value of observation i
 n = number of observations

As a secondary measure, Mean Absolute Error (MAE) is reported to provide a more robust assessment that is less sensitive to outliers:

$$MAE = \frac{1}{N} \sum_{i=1}^N |y_i - \hat{y}_i|$$

y_i = value of observation i
 $\hat{y}$ = predicted value of observation i
 n = number of observations

Finally, R^2 is used to assess the proportion of variance explained by the model. Negative values of R^2 indicate that the model performs worse than a simple mean prediction, serving as an important diagnostic for model validity in the test period.

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2}$$

y_i = value of observation i
 $\hat{y}$ = predicted value of observation i
 $\bar{y}$ = mean of observations
 n = number of observations

In addition, a computational complexity analysis is conducted by comparing training times across model classes, highlighting the trade-off between predictive performance and computational cost.

5 Results

5.1 RQ1 — Pooled Model Performance

The naïve persistence benchmark is strong precisely because HPI growth exhibits substantial autocorrelation (lag-1: 0.63), leaving limited additional signal for more complex models to exploit. As seen in Table 4, Ridge achieves the best pooled performance (RMSE=2.310, -12.1% vs. naïve), outperforming both benchmarks and tree-based methods. That regularised

linear models outperform Random Forest and Gradient Boosting suggests the dominant relationships in this panel are largely linear, consistent with Goulet Colombe et al. (2022), who find ML gains are largest when non-linearities are prominent. All models produce negative R^2 on the test period, honestly reflecting limited practical utility when out-of-sample dynamics diverge substantially from the training distribution. ML models provide statistically measurable but economically modest improvements; the most predictive signal is already captured by autoregressive dynamics.

Model	RMSE	MAE	R2	vs Naïve	vs OLS
Naïve	2,629	1,995	-0,437	-	-
OLS	2,366	1,819	-0,165	+10,0%	-
Rolling mean	2,331	1,739	-0,130	+11,3%	-
Ridge	2,310	1,775	-0,110	+12,1%	+2,4%
Lasso	2,342	1,799	-0,141	+10,9%	+1,0%
Random Forests	2,355	1,770	-0,153	+10,4%	+0,5%
Gradient Boosting	2,360	1,793	-0,158	+10,2%	+0,3%

Table 4: Pooled model performance on the test set (2022–2025) across three metrics: RMSE, MAE, and R^2 . Models are compared by RMSE.

A residual analysis of the pooled Ridge model, the best performing, confirms a mild systematic over-prediction bias: the mean residual is 0.91 percentage points indicating that the model tends to over-predict HPI growth on average in the 2022–2025 test period. This is consistent with the structural break that happened in the period post-2022: this pushed the test period outside the training distribution.

5.2 RQ2 — Stratified vs Pooled Comparison

Stratification improves accuracy selectively, and the clustering scheme matters more than model complexity. As seen on *Figure 13* literature-based stratification outperforms the pooled baseline in 6 of 8 combinations (average +11.56%). The North cluster delivers the strongest result in the study: GB achieves a 29.5% RMSE reduction (RMSE=1.494, R^2 =+0.337), the only positive R^2 across all model-cluster combinations. West (+14.3% GB) and East (+11.9% RF) follow. The South cluster deteriorates under stratification (RF -4.4%, GB -1.5%), consistent with its internal heterogeneity; Cyprus, Italy, Portugal and Spain follow fundamentally different trajectories despite sharing a geographic label. The pattern is consistent: within-cluster homogeneity drives stratification gains, and the literature typology is a reliable proxy for it when it reflects genuine institutional similarity, and an unreliable one when it does not.

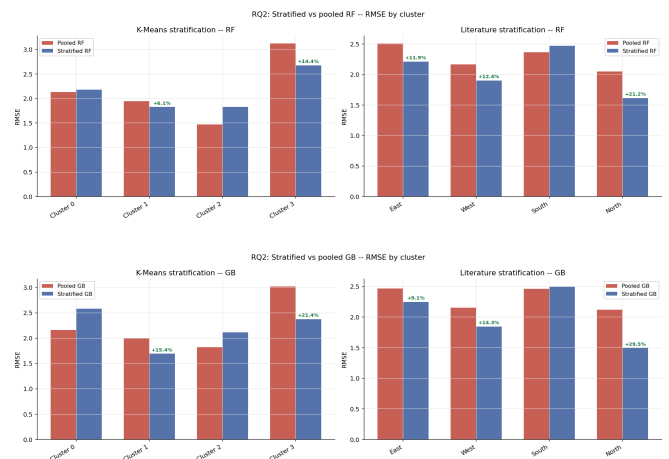


Figure 13: RMSE comparison for Random Forests (upper) and Gradient Boosting (bottom) methodologies throughout literature and KMeans clusters.

K-Means yields no aggregate gain (4 of 8 cases, average -0.63%). The Ireland-Malta cluster

produces severe deterioration (RF -24.5% , GB -16.2%), while Cluster 3 — high-growth convergence economies — achieves $+21.4\%$ GB, suggesting data-driven clustering can identify coherent regimes when structural separation is sharp. The overall K-Means underperformance reflects a fundamental asymmetry: it operates on macroeconomic variables, while the literature typology encodes institutional characteristics (mortgage structure, welfare regime, privatisation history) absent from the feature space, acting as external institutional supervision rather than a competing method on equal footing.

5.3 RQ3 — Feature Importance

Interest rate variables account for approximately 30% of total RF importance, with lagged HPI growth contributing a further 19%, confirming monetary sensitivity and short-term price persistence as seen on *Figure 14*.

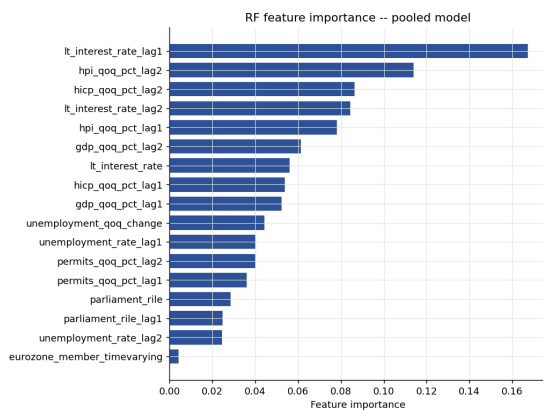


Figure 14: Random Forest features importance for the pooled model, measured by mean decrease in impurity.

Parliament ideology (RILE) shows non-trivial importance, supporting the inclusion of institutional variables, while eurozone membership contributes negligibly once interest rates are controlled for. Regime-specific analysis reveals meaningful variation:

West and East are primarily interest-rate driven; North is shaped by inflation dynamics; South shows a mixed structure. This regime-dependence of feature importance independently corroborates the stratified modelling approach, as a single pooled relationship would not produce such divergent regime-specific rankings.

5.4 Complexity and Running Time

The relationship between model complexity and computational cost is illustrated in *Figure 15* and complemented by the complexity–performance comparison (*Figure A5*). Training times increase substantially with model complexity: linear models (OLS, Ridge, Lasso) train almost instantaneously (≈ 0.01 – 0.02 seconds), while tree-based methods are an order of magnitude slower, with Random Forest and Gradient Boosting requiring approximately 0.28 and 0.46 seconds, respectively.

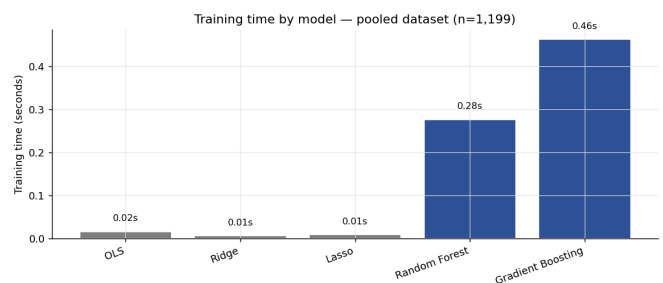


Figure 15: Running times of pooled models.

These results highlight a clear trade-off: more complex models incur significantly higher computational costs without proportionate gains in predictive accuracy.

6 Ethical Considerations

This study uses only aggregated public data from Eurostat, so no direct privacy concerns arise. However, national-level predictions mask within-country distributional effects (urban-rural divergence, impact on lower-income households) and should not be interpreted as representative of subnational markets or vulnerable groups. Additionally, observation lengths vary from 48 to 84 quarters per country, meaning models are trained on less evidence for recently acceded EU members; predictions for these countries should be interpreted with greater caution.

7 Limitations

Weak cluster structure: Silhouette scores remain below the conventional 0.5 threshold across all values of K tested, indicating that European housing markets form a structural continuum rather than discrete regimes. Ireland's East assignment rests on theoretical rather than empirical grounds — DBSCAN consistently classifies it as noise and K-Means groups it with Malta.

Exclusion of Greece: Greece was excluded due to structural data gaps, omitting one of the most extreme post-crisis housing market contractions in the EU-27.

Test period specificity: The 2022–2025 test set captures conditions that diverge substantially from the training distribution, and results may not generalise to more stable environments.

Panel imbalance: Observation counts range from 48 to 84 quarters per country, meaning recently acceded EU members are structurally underrepresented in the training data.

No sub-national decomposition: National-level HPI averages mask substantial within-country dispersion, particularly in large economies where capital city dynamics diverge sharply from national trends.

Feature importance interpretation: Impurity-based Random Forest importance in a pooled panel conflates between-country and within-country variation. Interest rate dominance should be interpreted as reflecting the most informative variable in the panel overall, rather than a precise causal within-country estimate.

8 Conclusion & Future Work

This paper forecasts one-quarter-ahead house price growth across 26 EU member states using a quarterly panel of 1,704 observations (2005–2025). The core finding is that model design choices matter more than model complexity. Ridge regression outperforms Random Forest and Gradient Boosting in the pooled setting, while literature-based regime stratification produces selective but meaningful gains — most strongly in the North cluster (GB: +29.5%, $R^2=+0.337$), the only positive R^2 in the study. Within-cluster institutional homogeneity is the operative condition for stratification success: the literature typology outperforms data-driven K-Means because it encodes institutional dimensions absent from the macroeconomic feature space. Where that typology reflects genuine structural similarity it helps; where it does not (South) it hurts. DBSCAN's inability to find more than one dense cluster confirms that European housing markets form a continuum, which explains why stratification gains are selective rather than universal.

Several directions remain open. Soft clustering via Gaussian Mixture Models would better capture the gradient structure that K-Means hard assignments obscure. Extending the panel to NUTS-2 level would address within-country heterogeneity invisible at the national level. Incorporating mortgage rate data for the Eurozone sub-panel and higher-frequency sentiment indicators could provide leading signals not captured in quarterly aggregates. Finally, a regime-switching approach allowing cluster memberships to evolve over time would better reflect the ongoing convergence of Eastern economies toward Western market characteristics.

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Figure A1: Boxplots of target by country and year.

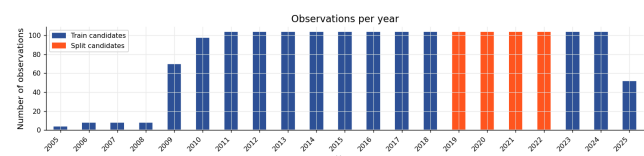
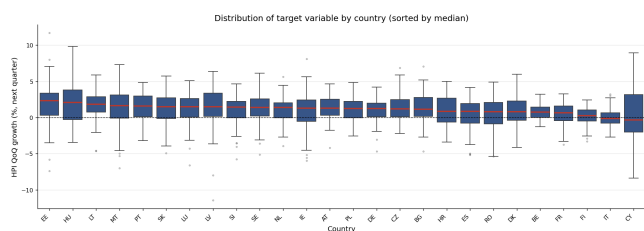


Figure A2: Barplot of different possible test-train splits, based on data percentage per split (Split at 2019: train=1028, test=676 (39.7% test); Split at 2020: train=1132, test=572 (33.6% test); Split at 2021: train=1236, test=468 (27.5% test); Split at 2022: train=1340, test=364 (21.4% test))

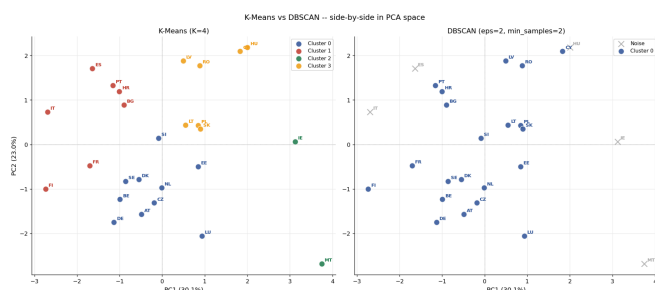


Figure A3: K-Means and DBSCAN based clustering in PCA space.

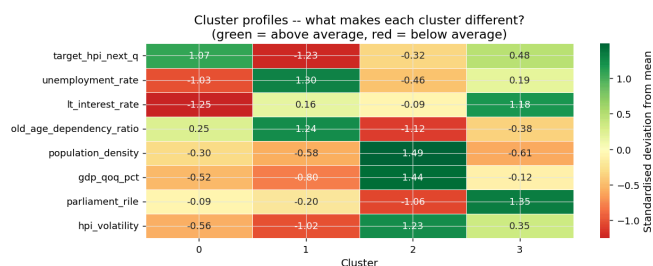


Figure A4: Clustering profiles heatmap.

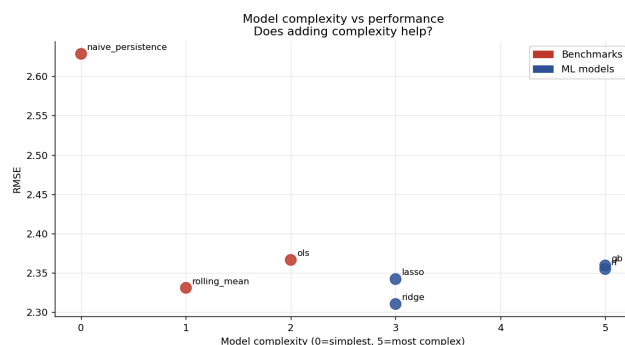


Figure A5: Model complexity vs. RMSE for pooled models

scheme	cluster	model	rmse_pooled	rmse_strat	improvement	n
K-Means	Cluster 0	GB	2.156775	2.577693	-19.516105	140
K-Means	Cluster 0	RF	2.129160	2.174660	-2.136988	140
K-Means	Cluster 1	GB	1.997082	1.689095	15.421844	98
K-Means	Cluster 1	RF	1.941981	1.822725	6.140937	98
K-Means	Cluster 2	GB	1.817737	2.111991	-16.187942	28
K-Means	Cluster 2	RF	1.468055	1.827502	-24.484545	28
K-Means	Cluster 3	GB	3.016746	2.372655	21.350534	98
K-Means	Cluster 3	RF	3.119856	2.671849	14.359845	98
Literature	East	GB	2.468128	2.244638	9.055022	168
Literature	East	RF	2.508265	2.210517	11.870684	168
Literature	North	GB	2.119568	1.493653	29.530312	42
Literature	North	RF	2.046945	1.613566	21.171977	42
Literature	South	GB	2.463134	2.499073	-1.459069	70
Literature	South	RF	2.363234	2.466958	-4.389054	70
Literature	West	GB	2.153914	1.845285	14.328742	84
Literature	West	RF	2.167415	1.899273	12.371506	84

Table A1: Models stratified comparison

Appendix B – Dataset Documentation

Project Overview

Country-quarter panel covering up to EU-27 members, spanning approximately 2005–2025 (~1,700 rows). Goal: forecast next-quarter residential house price growth (QoQ %) and compare models trained on data-driven vs. literature-based country clusters.

Datasets at a Glance

#	Dataset	Source	Raw Freq.	Role
1	House Price Index (HPI)	Eurostat <i>prc_hpi_q</i>	Quarterly	Target variable + autoregressive feature
2	HICP Inflation	Eurostat <i>prc_hicp_midx</i>	Monthly	Demand-side cost pressure
3	GDP Index	Eurostat <i>namq_10_gdp</i>	Quarterly	Economic activity
4	Unemployment Rate	Eurostat <i>une_rt_q</i>	Quarterly	Labour market conditions
5	Building Permits Index	Eurostat <i>sts_cobp_q</i>	Quarterly	Housing supply proxy
6	Old-age Dependency Ratio	Eurostat <i>demo_pjanind</i>	Annual	Structural demographics (clustering)
7	Population Density	Eurostat <i>demo_r_d3dens</i>	Annual	Urbanisation / land scarcity (clustering)
8	MIR Mortgage Rates	ECB API	Monthly	Cost of borrowing — eurozone only
9	Long-term Interest Rates	Eurostat <i>irt_lt_mcby_q</i>	Quarterly	Sovereign yields — all EU-27
10	Manifesto Project (MARPOR)	MARPOR (2025a) <i>mpd_data.csv</i>	Per election	Parliament ideology score

Variable Notes by Group

1. Housing Market — Target & Momentum

House Price Index (HPI) — Nominal index (2015 = 100). Used in two forms: QoQ % change as a momentum feature (*hpi_qoq_pct*), and shifted one quarter forward as the prediction target (*target_hpi_next_q*). QoQ growth is preferred over the index level for stationarity. The index clearly captures the 2008–09 crash, the post-2015 recovery, the COVID-era boom, and the 2022–23 correction driven by ECB rate hikes.

2. Macroeconomic Indicators

Harmonized Consumer Price Index (HICP) — Quarterly mean of monthly observations. Affects housing through multiple channels simultaneously: real income erosion, construction cost inflation, and ECB tightening signals. The direction of its net effect on prices is empirically ambiguous.

Gross Domestic Product (GDP) — Chain-linked volume index (2005 = 100), seasonally and calendar adjusted. QoQ growth measures short-term economic momentum.

Unemployment — ILO definition, seasonally adjusted, ages 15–74. Both the level and QoQ change are included: a country with 8% unemployment but falling is structurally different from one at 8% and rising.

3. Supply-Side Constraints

Building Permits — Index (2021 = 100), residential dwellings only. A leading supply indicator: permits precede actual completions by 6–18 months, so current permit volumes partly predict supply conditions two to four quarters ahead.

4. Financing Costs

MIR Mortgage Rates — Actual annualised agreed rate on new euro-denominated mortgage loans. Structurally absent (NaN) for 7 non-eurozone countries: DK, SE, PL, CZ, HU, BG, RO.

Long-term Government Bond Yields — 10-year government bond yields (EMU convergence criterion). Captures sovereign risk and benchmark for fixed-rate mortgage pricing. Crucially provides an interest rate signal for all EU-27.

5. Structural, Demographic, and Political Features

Old-age Dependency Ratio — People aged 65+ per 100 working-age people. Used primarily as a clustering feature.

Population Density — Inhabitants per km². Proxies for land scarcity and urbanisation intensity.

Parliament RILE (MARPOR) — Seat-weighted average of party right-left scores across the full legislature. Included at level and two lags.

Panel Construction

Frequency Harmonisation: Monthly series (HICP, MIR) converted to quarterly means. Annual demographics carried forward. Election data carried forward until the next election.

Missing Data Treatment

Variable	Gap	Treatment
<i>mortgage_rate_pct</i>	Absent for 7 non-eurozone countries	Retained as NaN
<i>lt_interest_rate</i>	~44 values (Malta, Cyprus early)	Forward-filled
<i>parliament_rile</i>	NaN before first recorded election	Good coverage from 2005
Demographics	Sparse early-year gaps	Forward-filled

Final Panel: ~1,700 rows after dropping missing target observations. 26 countries $\times$ 65–80 quarters.

Appendix C – Contribution & LLM Usage Disclosure

Component / Module	Contributors	LLM	Type of Usage	% LLM	Validation / Modification
Data pipeline (data_exploring.ipynb)	Carla, Christoph	Claude (Sonnet 4.5)	Debugging, code generation, structural improvements and cleaning.	25%	Tested on full dataset; manual review of all filter logic
EDA (eda.ipynb)	Carla, Christoph,	Claude (Sonnet 4.5)		25%	All interpretations written by us
Clustering (clustering.ipynb)	Carla, Giulia	Claude (Sonnet 4.5)		35%	Cluster assignments manually validated against country profiles
Modelling (modelling.ipynb)	Carla, Christoph, Giulia	Claude (Sonnet 4.6)		35%	Hyperparameter grid manually specified; results independently verified
Metrics (metrics.ipynb)	Carla, Giulia	Claude (Sonnet 4.6)		25%	All metric interpretations written by us
Report Writing	Carla, Christoph, Giulia	Claude (Sonnet 4.5)	Structure suggestion; report correction (Prompt: “Imagine you are the professor, considering the course guidelines evaluate this report and give a grade”).	20%	Report rewritten and modified based on corrections and suggestions (for example, poor writing structure corrections, language improvement).
Additional Resources	Christoph				